

Integrating GIS, R, and a Digital Sewage Capacity Model for Flood Reconstruction and Simulation: Insights from the Brooklyn October 2025 Flood

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Abstract

In recent years, Brooklyn has experienced several floods that suggest the borough's stormwater infrastructure is insufficient. However, this flooding could also be caused by the landscape, so it's important to evaluate the primary cause of this issue. The study area was determined by analyzing background signage in videos and images of the NYC October 2025 flood and pinning them in GoogleEarth. Using GIS data sets to measure the length of sewer pipes and sewer design standards from the NYC Department of Environmental Protection, the capacity of the sewage system was calculated and sea-rise flooding was simulated. The simulations showed that the flooding patterns do not match that of a landscape flood, and that floods occurred even in areas with larger sewage capacity like on major avenues. Sewage capacity was relatively consistent across the borough, meaning capacity is not a significant issue. This then suggests flooding in multiple study points was not caused by low-lying elevation nor insufficient capacity, but by clogging of particular areas. However, data values were estimated and rainfall flooding simulation was unavailable. By analyzing these flood causes, this study could help identify courses of action to reduce future flood events in Brooklyn and provide a method for other cities to analyze the causes of their own flood-prone areas.

Introduction

The New York City sewer system is designed to handle 1.75 inches of rain per hour (New York City Mayor's Office of Resiliency, 2021). Similar increases in flood frequency and intensity have been documented in both coastal and urban environments, where climate-driven precipitation changes and evolving hydrologic conditions amplify existing infrastructure limitations (Tabarus et al., 2020; Mead et al., 2023; Tsakiri et al., 2018). More specifically, the size and structure of sewers across the city are based on a precipitation intensity-duration-frequency (IDF) curve that is based on historical rainfall and runoff data from 1903-1951, which means that the sewers are designed to effectively drain away runoff from worst-case scenario storms that were historically assumed to only occur every 5-years in New York City (New York City Mayor's Office of Resiliency, 2021). However, recent studies suggest that such design assumptions are increasingly insufficient, as urban flooding is influenced not only by rainfall intensity but also by drainage system performance and localized infrastructure

constraints (Marsellos et al., 2016; Lakeram et al., 2017). Increased rainfall rates have been observed globally and largely attributed to climate change (Westra et. al, 2013). In recent years, New York City has experienced near record-levels of rainfall more often than every 5 years (Aggarwala, 2024), and its sewer system has not handled it well, as demonstrated by intense flooding observed during these events. This mismatch between expected and observed flood behavior highlights the need to evaluate both topographic controls and infrastructure performance when assessing urban flooding mechanisms.

On September 29, 2023, 5.48 inches of rain were recorded in Central Park (National Weather Service, 2026), and on October 30, 2025, 1.83 inches were recorded (National Weather Service, 2026). Many people took to social media to share videos of knee-deep flooding across Brooklyn during these events. A previous study in 2024 (Lawlor et al., 2024) analyzed the September 2023 flood using flood video analysis, Google Earth, and Global Mapper to map vulnerable areas and their proximity to emergency services. Similar GIS-based methodologies have been widely applied in flood studies using LiDAR-derived Digital Elevation Models (DEMs) and volumetric simulations to reconstruct flood events and evaluate inundation patterns (Marsellos et al., 2010; Foster et al., 2011; Foster & Marsellos, 2012). These studies served as inspiration for this one, where the concept was taken further to investigate the underlying causes of Brooklyn's severe flooding. Using similar methods, this study aims to determine whether Brooklyn's floods are primarily driven by landscape conditions or sewer system behavior, recognizing that flood processes are influenced by both topographic variability and infrastructure limitations such as hydraulic connectivity and flow constraints (Marsellos et al., 2016; Tsakiri et al., 2018). In riverine systems, for example, obstacles such as bridges, dams, and localized blockages have been shown to significantly alter water flow and increase flooding upstream, emphasizing that non-topographic controls can play a dominant role in flood propagation (Foster et al., 2011; Lakeram et al., 2017).

This study builds upon prior GIS and LiDAR-based methodologies by extending the analysis beyond terrain-driven flooding to include infrastructure-based factors, particularly sewer system capacity and performance. By combining flood reconstruction, elevation-based simulation, and a Digital Sewage Capacity Model (DSCM), this work aims to distinguish between topographic and infrastructure-driven flooding processes, contributing to improved urban flood risk assessment and management strategies.

Methodology

We collected images and videos of the flooding event from multiple media outlets, identifying 17 different geographic locations. This data was sourced from YouTube, social media platforms, and news outlets. The publication date of each source was verified to ensure it

corresponded to the specific flood event examined in this study. Locations were determined by analyzing background features, such as street signage and building names, to confirm that the footage was captured in Brooklyn and to identify specific sites. Water depth at each location was estimated by comparing observed water levels in the images to standard dimensions of urban features, including vehicle tires, streetlamp bases, and curbs (see Table 1 for all locations, estimated depths, associated uncertainties, and links to source media). Given the variability in camera angles, perspectives, and image quality, these measurements are subject to uncertainty. Accordingly, an uncertainty range was assigned to each depth estimate. This variability reflects the lack of standardized measurement conditions across the collected media. All identified locations were plotted in Google Earth Pro, where a perimeter encompassing the study area was delineated. The resulting boundary formed a roughly triangular-shaped polygon that encompasses the selected data points across Brooklyn, solidifying the “study area” examined during this experiment (Figure 1). This polygon was then exported as a .kmz file and subsequently imported into Global Mapper (version 26.1) for further analysis. Additional datasets were incorporated into Global Mapper to produce a more complete representation of Brooklyn. These included a World Imagery Projection, the USGS 3D Elevation Program (3DEP) Projection, and a detailed road network layer. The USGS 3DEP Projection was selected because it is a bare-earth, 1-meter LiDAR digital elevation model (DEM), which provides accurate elevation measurements without interference from buildings or vegetation. This is important for flood analysis, as ground elevation is a primary factor influencing flood extent and water accumulation. After loading the initial study area, vector boundaries were overlaid and adjusted to create a more rectangular perimeter better suited for grid-based analysis. The revised study area encompassed approximately 200 square kilometers. This updated study area was then used as the basis for subsequent flood analysis.

Table 1: Locations of observed flooding across Brooklyn during the October 2025 storm event, including geographic coordinates, estimated water depths with associated uncertainties, and corresponding source media used for reconstruction.

Map #	Location (Intersection, Address, Neighborhood)	Coordinates (Longitude, Latitude)	Water Height Inferred (cm) & Uncertainty ($\pm$)	Link to Source and Timestamp/Note*
0	4024 Farragut Rd	40°38'11.91" N	18 $\pm$ 2 cm	https://youtu.be/1ywSNVWsQII?si=N3xj78fEb3VLCKR4
	East Flatbush	73°56'15.62" W		
1	1523 Avenue M	40°37'06.48" N	12 $\pm$ 3 cm	https://www.youtube.com/watch?v=jgXVxQ3wo1s
	Midwood	73°57'33.65" W		

2	372 4th Avenue Park Slope	40°40'22.68" N 73°59'12.52" W	18 ± 2 cm	https://www.tiktok.com/@sosyy122/video/7567397149529378103?is_from_webapp=1&sender_device=pc&web_id=7608235670612231711 (00:25-00:40)
3	1933 Flatbush Ave Flatbush	40°37'22.23" N 73°56'11.58" W	8 ± 3 cm	https://www.tiktok.com/@unmainstreamed/video/7567833535868964151?is_from_webapp=1&sender_device=pc&web_id=7608235670612231711 (00:08-00:46)
4	4th Ave, 5th Street Park Slope	40°40'21.93" N 73°59'12.02" W	18 ± 2 cm	https://www.youtube.com/shorts/TjY5qH109sU (full video)
5	6th Ave & Pacific St Prospect Heights	40°40'53.96" N 73°58'26.12" W	28 ± 3 cm	https://www.tiktok.com/@squatsnsplits92/video/7567107534989839646?_r=1&_t=ZP-941nrvtOBdn (full video)
6	Knickerbocker Ave, Hart Street Bushwick	40°42'05.56" N 73°55'23.28" W	23 ± 2 cm	https://ny1.com/nyc/all-boroughs/weather/2025/10/30/death-after-flooding-as-city-sees-heavy-rain (01:09-01:12)
7	Lincoln Rd, Ocean Ave Prospect Park	40°39'37.92" N 73°57'46.03" W	5 ± 2 cm	https://www.youtube.com/shorts/QSDSj28iYbM (full video)
8	319 Knickerbocker Ave Bushwick	40°42'05.19" N 73°55'22.02" W	18 ± 2 cm	https://www.theguardian.com/us-news/video/2025/oct/31/heavy-rain-floods-streets-in-new-york-city-video (00:11-00:15)
9	8106 5th Ave Bay Ridge	40°37'29.90" N 74°01'29.48" W	22 ± 2 cm	https://youtu.be/QZT7G7BZ-9c?si=5JlVhmGj7VOakBmi (00:57-01:05)
10	830 Lafayette Ave Bed-Stuy	40°41'27.74" N 73°56'21.77" W	22 ± 2 cm	https://www.youtube.com/watch?v=k-qiygBA-7Q (03:33-04:10)
11	293 Grand Ave Clinton Hill	40°41'14.85" N 73°57'44.45" W	12 ± 3 cm	https://www.thecity sponge.com/p/oct30-storm (first image in article, first clip in gif)

12	1459 Fulton St Bed-Stuy	40°40'48.50" N 73°56'33.99" W	48 ± 3 cm	https://www.nydailynews.com/2025/10/30/flooding-swamps-brooklyn-stranding-commuters-storm-whips-through-nyc-new-jersey/ (7th & 8th images)
13	565 Throop Ave Bed-Stuy	40°40'57.29" N 73°56'26.64" W	45 ± 5 cm	https://youtu.be/g14es9PyKck?si=gRfn_b1H_BH9i-Vh (00:08-00:12)
14	1422 Fulton St Bed-Stuy	40°40'47.36" N 73°56'37.82" W	15 ± 5 cm	https://youtu.be/g14es9PyKck?si=gRfn_b1H_BH9i-Vh (00:02-00:04)
15	1466 Dekalb Ave Bushwick	40°42'04.00" N 73°55'20.58" W	45 ± 5 cm	https://youtube.com/shorts/XPXfhXkK92g?si=vuhtfnAf11YNEbJo (full video)
16	548 Rutland Rd Crown Heights	40°39'35.67" N 73°56'33.19" W	58 ± 3 cm	https://www.nytimes.com/interactive/2025/11/25/nyregion/nyc-storm-hurricane-flooding.html (first clip in article)

**Video and social media sources (e.g., YouTube, TikTok) were used exclusively for flood depth estimation and geolocation verification and are not included in the formal reference list.*

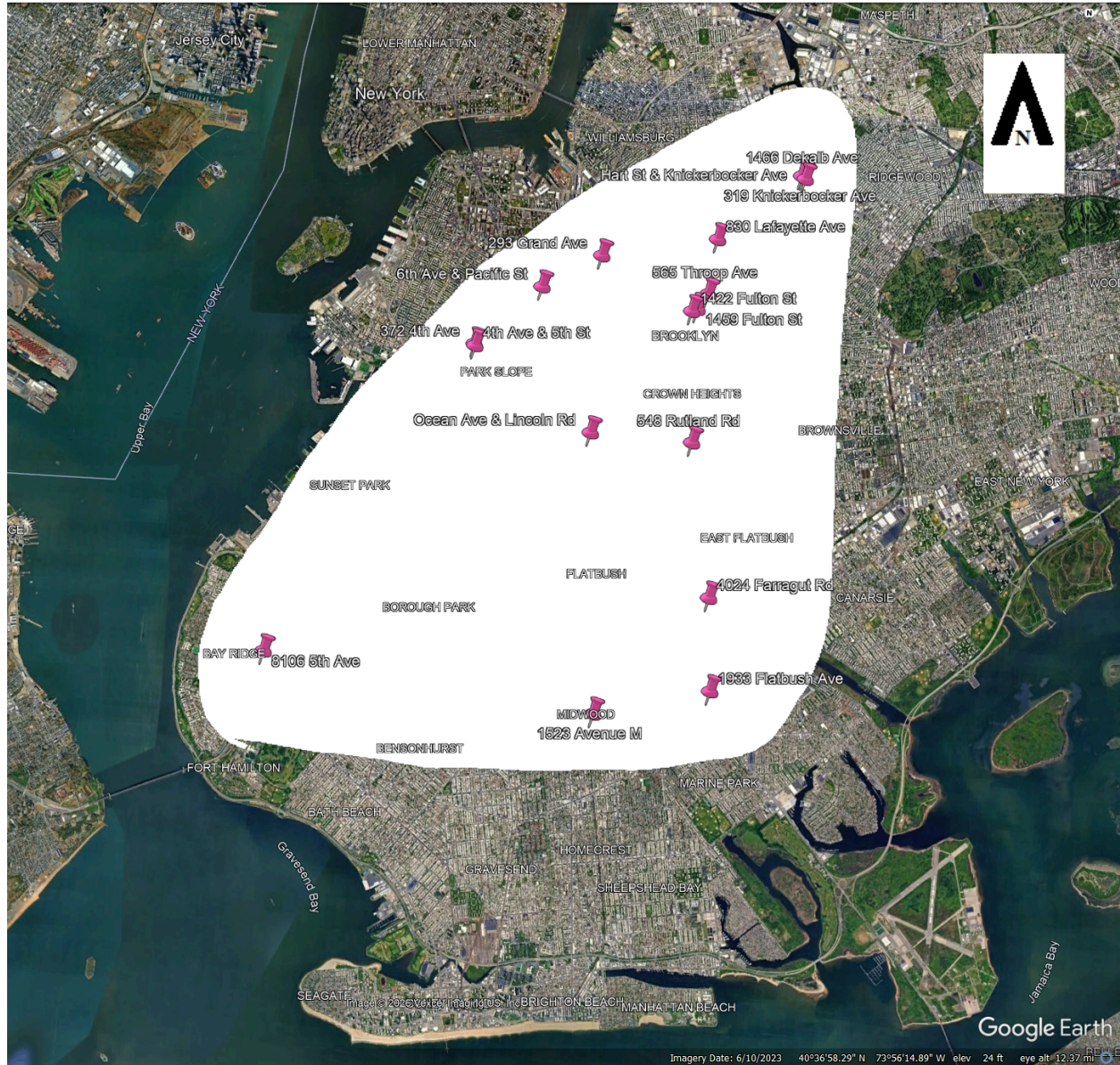


Figure 1: Initial study area parameters, designed to best encapsulate all data points as well as a decent area around them.

A key assumption of this study was that each road corresponds to an equivalent length of sewer pipe running parallel beneath it, and that pipe diameters are uniform for each road type. This assumption was necessary because the sewer infrastructure varies across different areas of the borough. For example, some neighborhoods use combined sewer systems which carry both storm and wastewater, while others use separate systems (New York City Department of Environmental Protection, 2025). Two road categories were considered in this study: local streets and major avenues. A pipe diameter of 0.508 meters was assigned for local streets, while major avenues were assigned a diameter of 0.914 meters. These values were based on commonly used pipe sizes used in sewer construction for these road types (New York City Department of Design

and Construction, 2025). Sewer pipes connected to individual buildings were excluded from this analysis, as building-level plumbing systems vary significantly and are not standardized. Roads within the study area were converted into a grid and assigned length values using Global Mapper functions. Each grid cell initially contained multiple road segments with individual length values. These values were aggregated so that each cell contained a single value representing the total road length within that cell. The attribute table of the gridded road dataset was then exported as a .csv file for statistical analysis in RStudio.

Once the dataset was imported into RStudio, several packages were utilized to process and modify the .csv file, which was converted into an R Data Frame. The “data.table” and “dplyr” packages were used for efficient data manipulation and aggregation. The “stringr” package was used to convert character values in the .csv file to numeric format, enabling mathematical operations. The “crayon” package was used for debugging, and the “foreign” package was used to convert the .csv file into a .dbf format for integration with shapefiles, which was the preferred table format in Global Mapper. Road length values within each grid cell were aggregated and then used to estimate sewer capacity. Specifically, the total pipe volume within each cell was calculated using the cylinder volume formula, with pipe length representing height and pipe radius based on the assigned road type. This calculated volume represents the estimated sewage capacity for each cell. It was assumed that when rainfall accumulates and discharge is insufficient (e.g., due to limited capacity or blockages), the sewer system reaches capacity, resulting in surface-level flooding.

Two road categories were analyzed: local streets and major avenues. Each category was processed separately, with individual .csv files generated and modified for each dataset (Figure 2). Using the gridded shapefile containing sewage capacity values (in cubic meters) for each cell, centroids were generated to represent the center point of each grid cell. These centroid points served as spatial representations of the estimated sewage capacity within each cell. The resulting grid was then converted into an elevation-style surface, where sewage capacity values were used in place of elevation. This model is referred to as the Digital Sewage Capacity Model (DSCM). The DSCM highlights areas of relatively high and low sewage capacity and enables comparison with previously identified flood locations. The final step involved flood simulation to assess whether observed flooding patterns were more consistent with topographic (landscape-driven) or infrastructure-related (sewer system) causes. Landscape-driven flooding would be expected to concentrate in areas of lower elevation, whereas sewer-related flooding would likely exhibit a more irregular spatial distribution with weaker correlation to elevation.

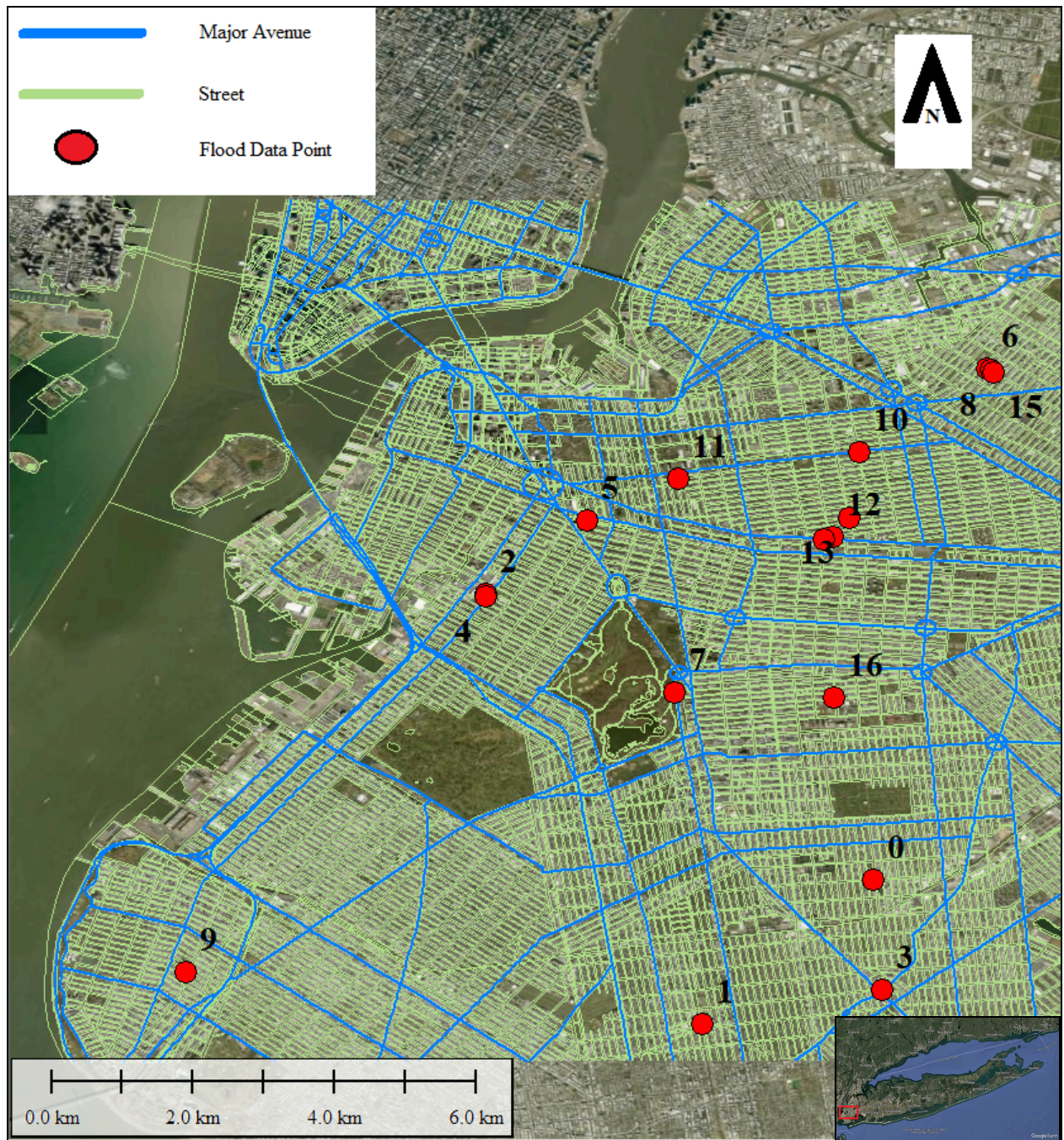


Figure 2: All roads in Brooklyn that fall within the specified study area. Major avenues are specified blue, and streets are specified in green. Roads/bridges in the small Manhattan portion are not considered in mapping analysis. Source: New York City Department of City Planning, 2025

Results

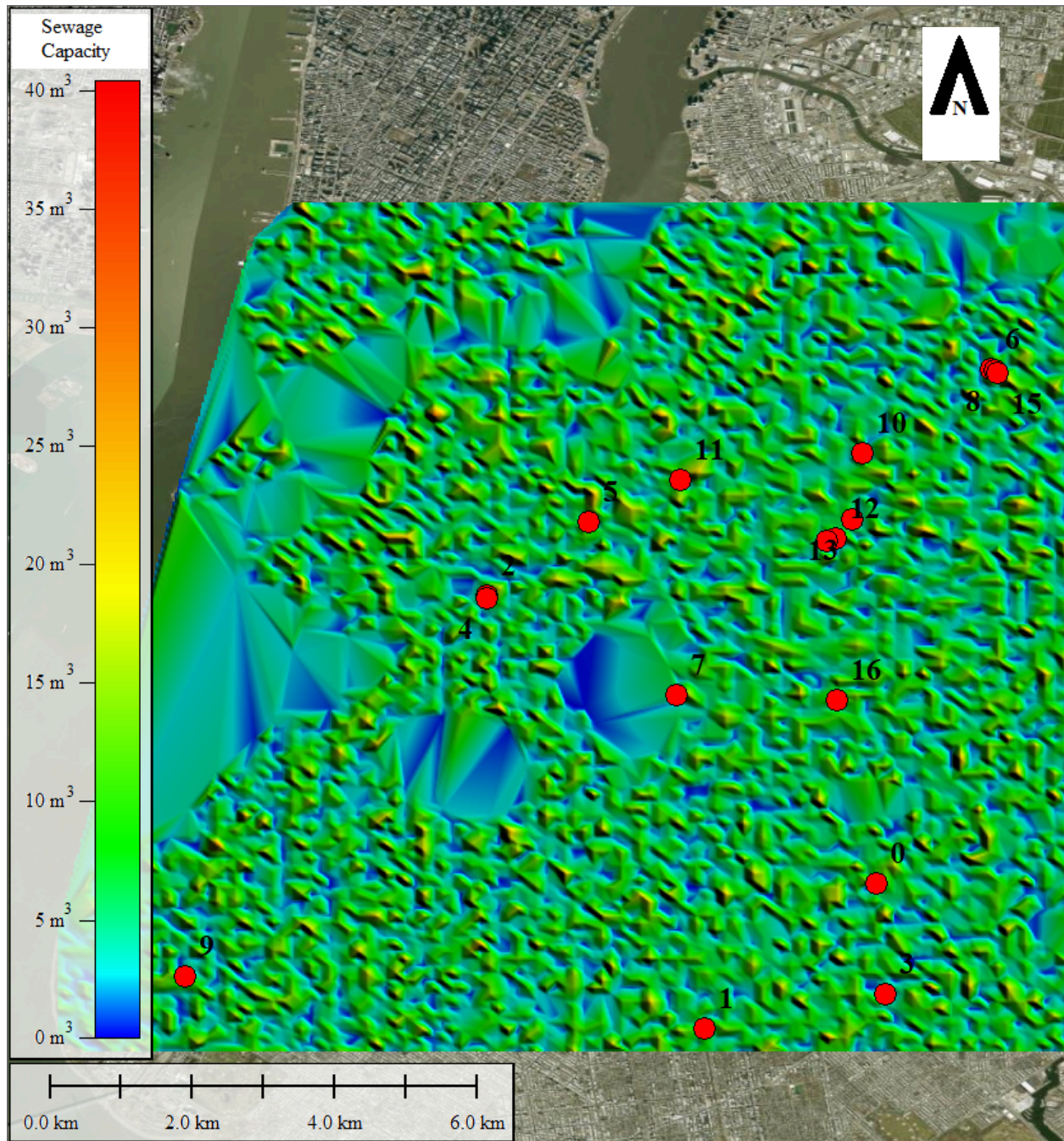


Figure 3a: Sewage capacity of Brooklyn streets represented as a colored map, referred to as a digital sewage capacity model. Large bluish-green voids can be seen, these represent parks and the local waterways.

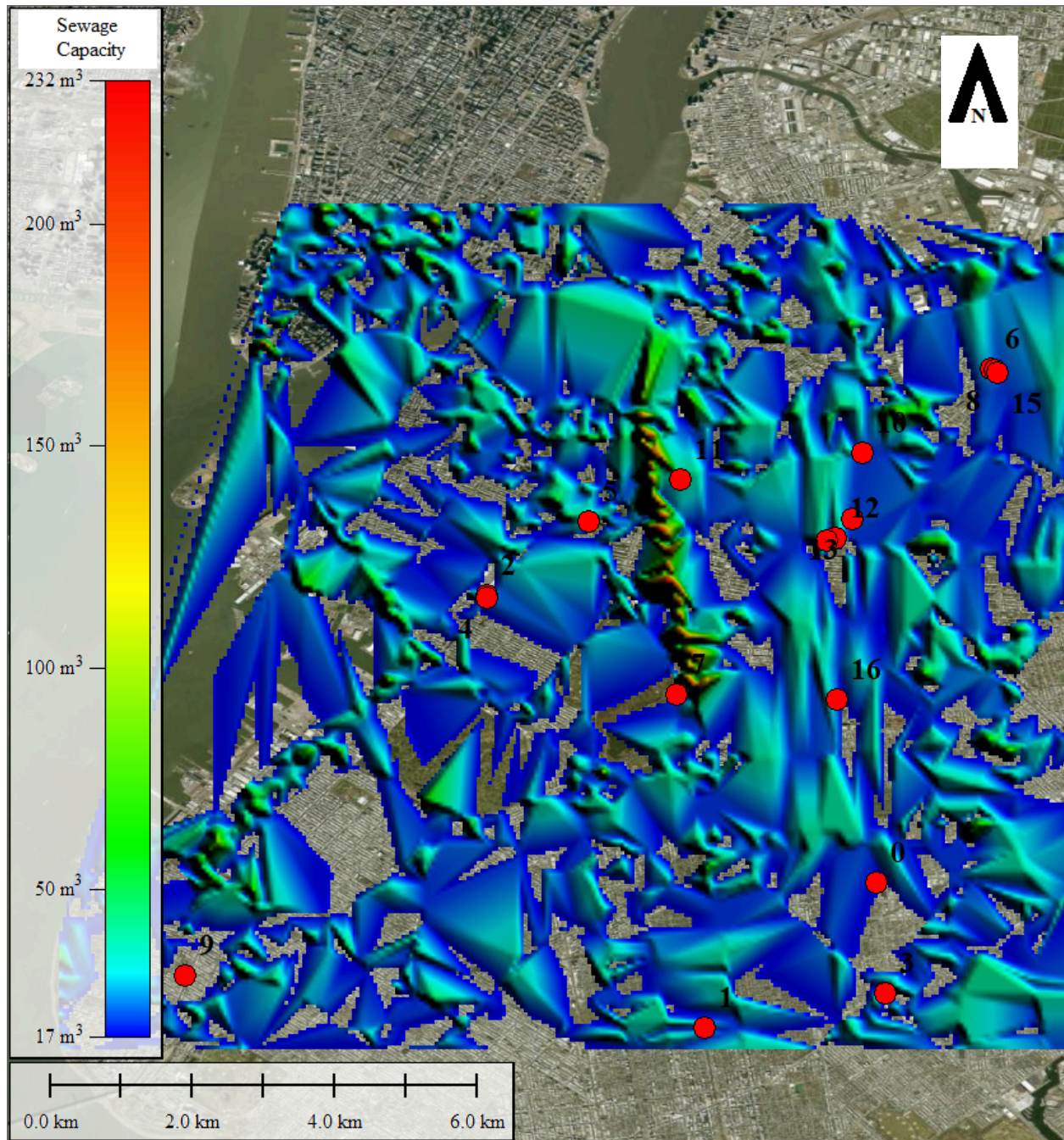


Figure 3b: Sewage capacity of Brooklyn’s major avenues, represented as a digital sewage capacity model. A smaller minimum scale was used because the average sewage capacity is larger and it excludes areas where only streets are present. Also, large gaps between major avenues tend to distort the look of the map. Some distortion is still present.

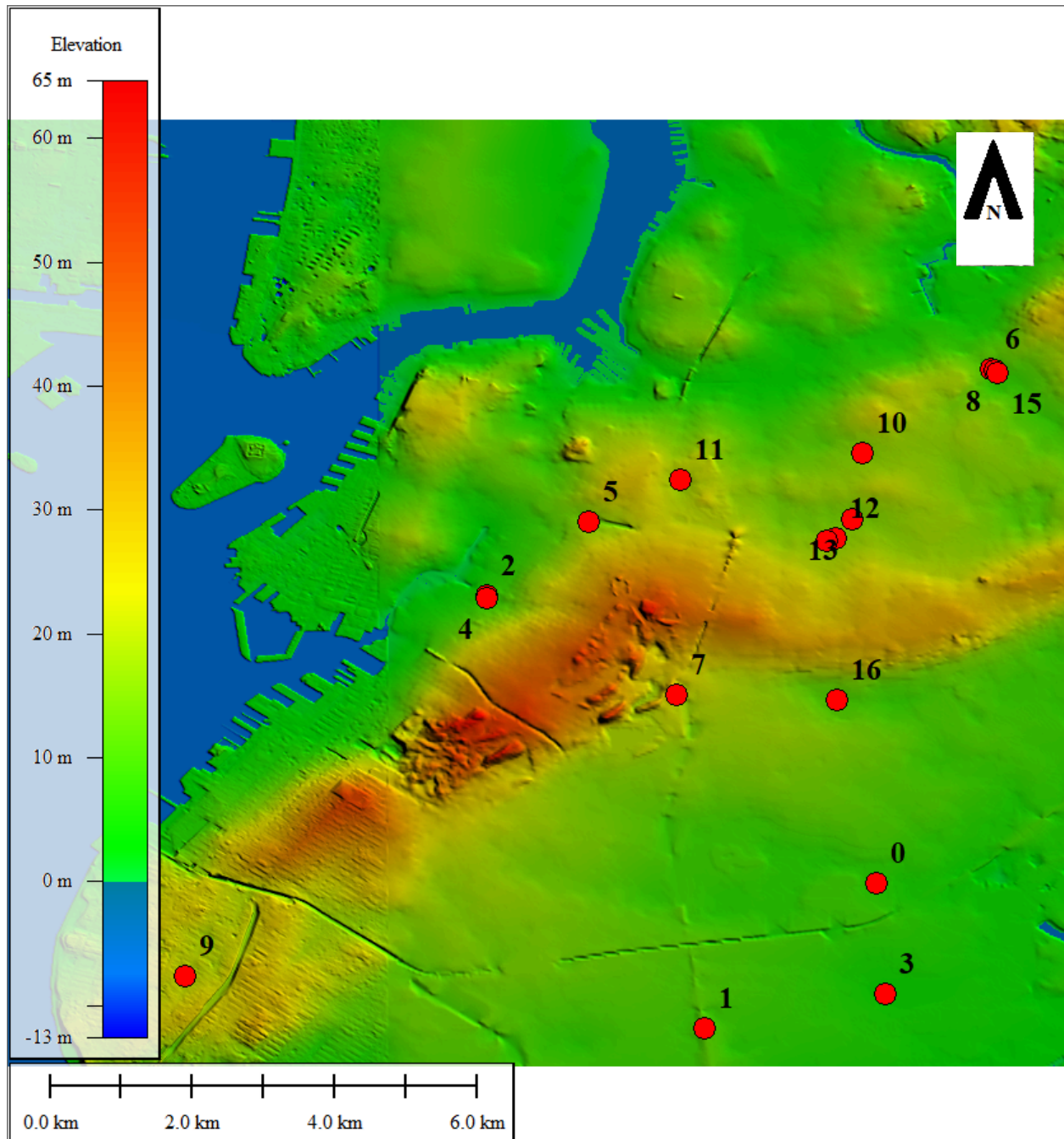


Figure 4a: Simulation of a landscape flood, with this view being the default DEM view of Brooklyn. Note the high elevation areas and the notable amount of recorded data points near them. Sea level is at default currently.

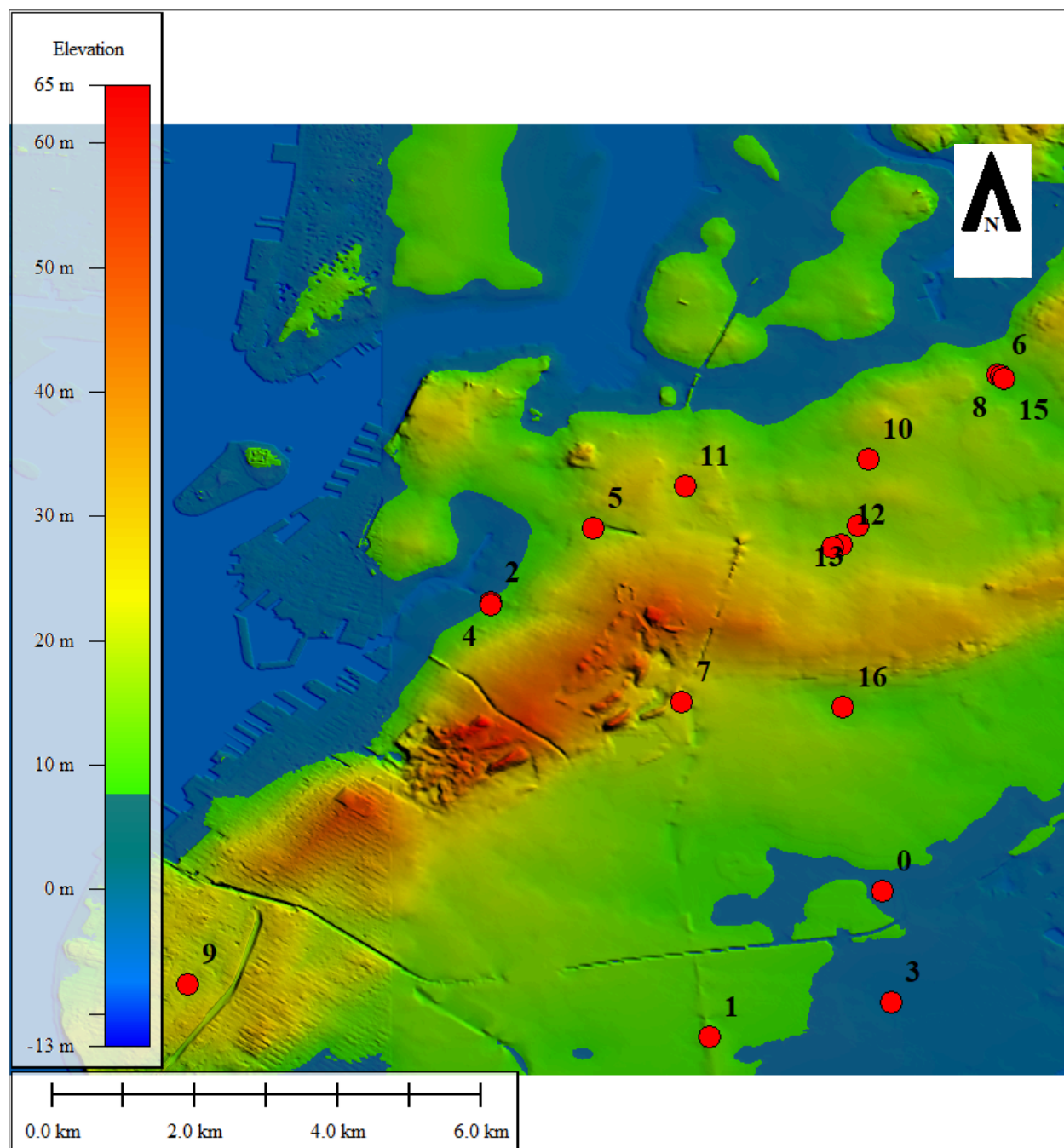


Figure 4b: Simulation of a landscape flood, but the water level has risen by 8 meters. Four data points have become submerged.

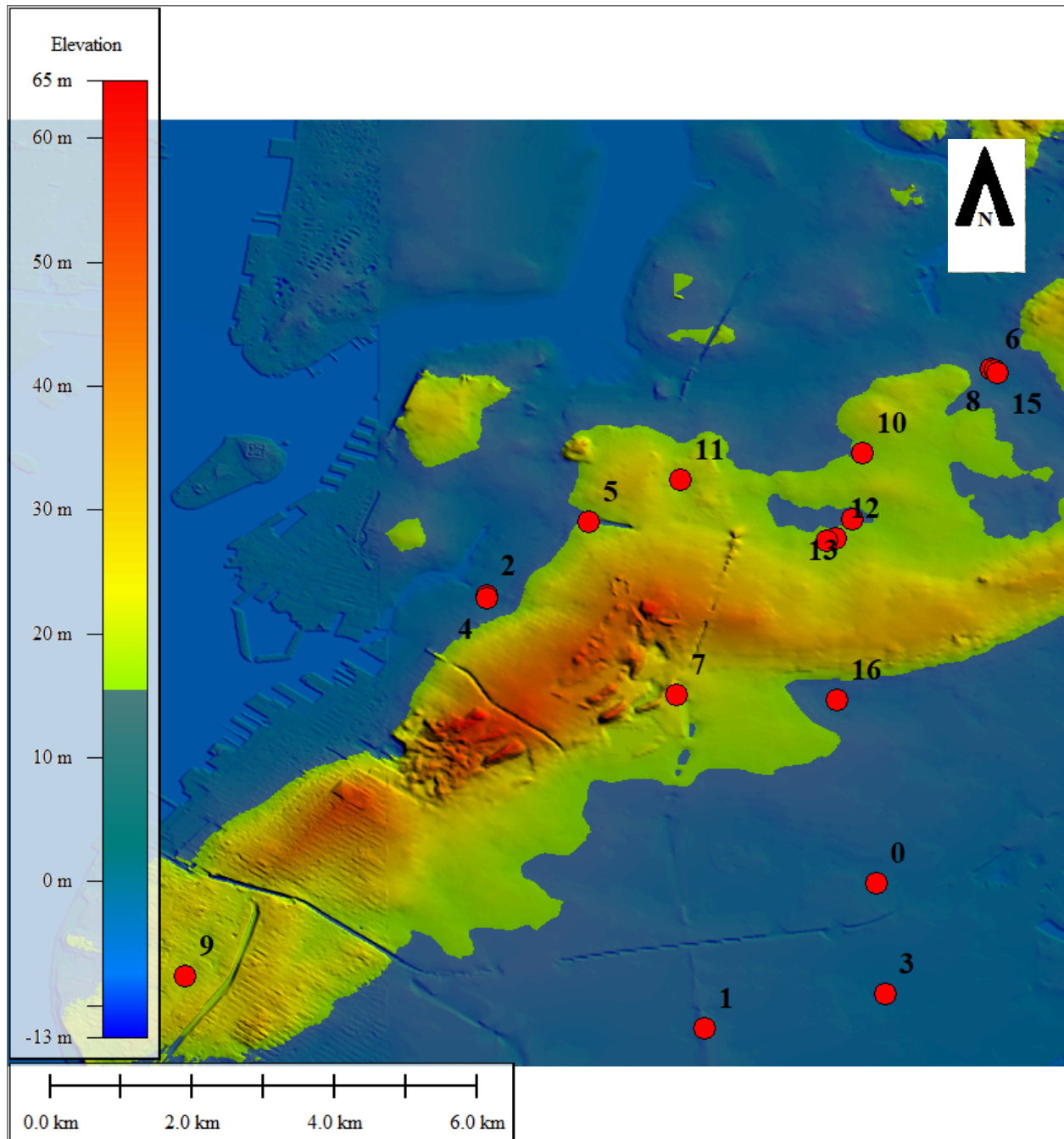


Figure 4c: Simulation of a landscape flood, but the water level has risen by 16 meters. Most of the flooded points have been overlapped by the water level, but some still are not.

The first two sets of figures detail sewage capacity in the streets and major avenues of Brooklyn. High capacity areas like Point 7 in Figure 3b are still victim to flooding. The distribution of sewage capacity across Brooklyn is relatively uniform, with expectedly more being present on major avenues (Figure 3a, 3b). The landscape flood simulation (Figure 4a, 4b,

4c) covered many points, but not all of them. At the end of this simulation, close to half the points still remain outside the flood zone.

Discussion

The results of this study suggest that sewer system limitations, rather than topography, were the primary driver of flooding during this event. The landscape-based flood simulation indicated that even under an extreme and unrealistic scenario, such as a 16-meter flood depth across Brooklyn, the modeled flood extent did not match the observed flooded locations identified earlier in this study. This is significant because only around 0.045-0.076 meters (1.8-3 inches) of rain actually hit the New York City area during this time (Spectrum News NY1, 2025; National Weather Service, 2026). A significant limitation to note is Global Mapper's inability to simulate a non-landscape flood, especially with uneven terrain not in a "bathtub" shape, such as Brooklyn. Given the distribution of flooded points and its poor matching to landscape flood patterns (Figure 4a, 4b, 4c), the flood was likely caused by a sewage issue.

Looking at the DSCM, the sewage capacity distribution across Brooklyn is relatively uniform, with most areas being consistently around 5-15 cubic meters in volume (Figure 3a). The major avenues have a much larger and variable capacity, and floods only occurred near major avenues once. This likely means the sewage discharge rate to waterways/waste treatment facilities was likely adequate for this flood in particular, suggesting that sewer backups could be the issue instead. Brooklyn received hundreds of reports to clean up grease in 2025, showing that this is an issue. Given the high request rate for sewer unblocking, the already present waste in sewer pipes likely clogs quickly when stormy conditions are present, since it effectively lowers capacity for a pipe (New York City Department of Environmental Protection, 2025).

Conclusion

Brooklyn's storm infrastructure issues appear to have been improving, the number of backup requests between 2023 and 2025 has fluctuated but is currently much lower, and the remaining backup requests have less to do with storms. It started with 52% stormwater-related backups in Fall 2022/Spring 2023, up to 70% in Fall 2023/Spring 2024, and down to 2% in Fall 2024/Spring 2025 (New York City Department of Environmental Protection, 2024; 2025a; 2025b). It should be noted that these numbers haven't caught up yet with the fall rainstorms of 2025, so further data would be needed to show definite improvement in this area.

While the study concludes that the Brooklyn flooding is not a landscape issue but rather a sewage clogging issue, many flaws in the study design must be addressed. The assumption that

applied to standardizing pipe diameter and length with the surface roads could be elaborated on further with better knowledge of the sewage system to produce a more accurate digital sewer capacity model; though this would still be challenging with the varying ages of different parts of the system. Further considerations could be taken to simulate the flood using software that allows for using rain on uneven terrain, something Global Mapper was unable to do. A more detailed capacity map that can detail discharge rates could be made by including water treatment plants, which affect the discharge rate of combined sewer outflow areas in certain sections of the borough (New York City Department of Environmental Protection, n.d.). These models demonstrate strong potential to accurately assess sewer capacity across different areas of the borough and identify locations most vulnerable to flooding. This represents an important step toward improving New York City's stormwater infrastructure.

Credit Authorship Contribution Statement

Persaud, D.: Data Collection, Global Mapper, programming in R (RStudio), Figures, Writing: Abstract, Methodology, Results, Discussion, Conclusion; Moore, O.: Data Collection, Research, Writing: Editing, Abstract, Introduction, Methodology, Discussion, Conclusion; Mercaldo, D.: Data Collection, Google Earth, RStudio, Microsoft Excel, References, Editing, Figures, Writing: Abstract, Introduction, Methodology, Discussion, Conclusion; Marsellos, A.E.: Supervision, Editing, RStudio

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